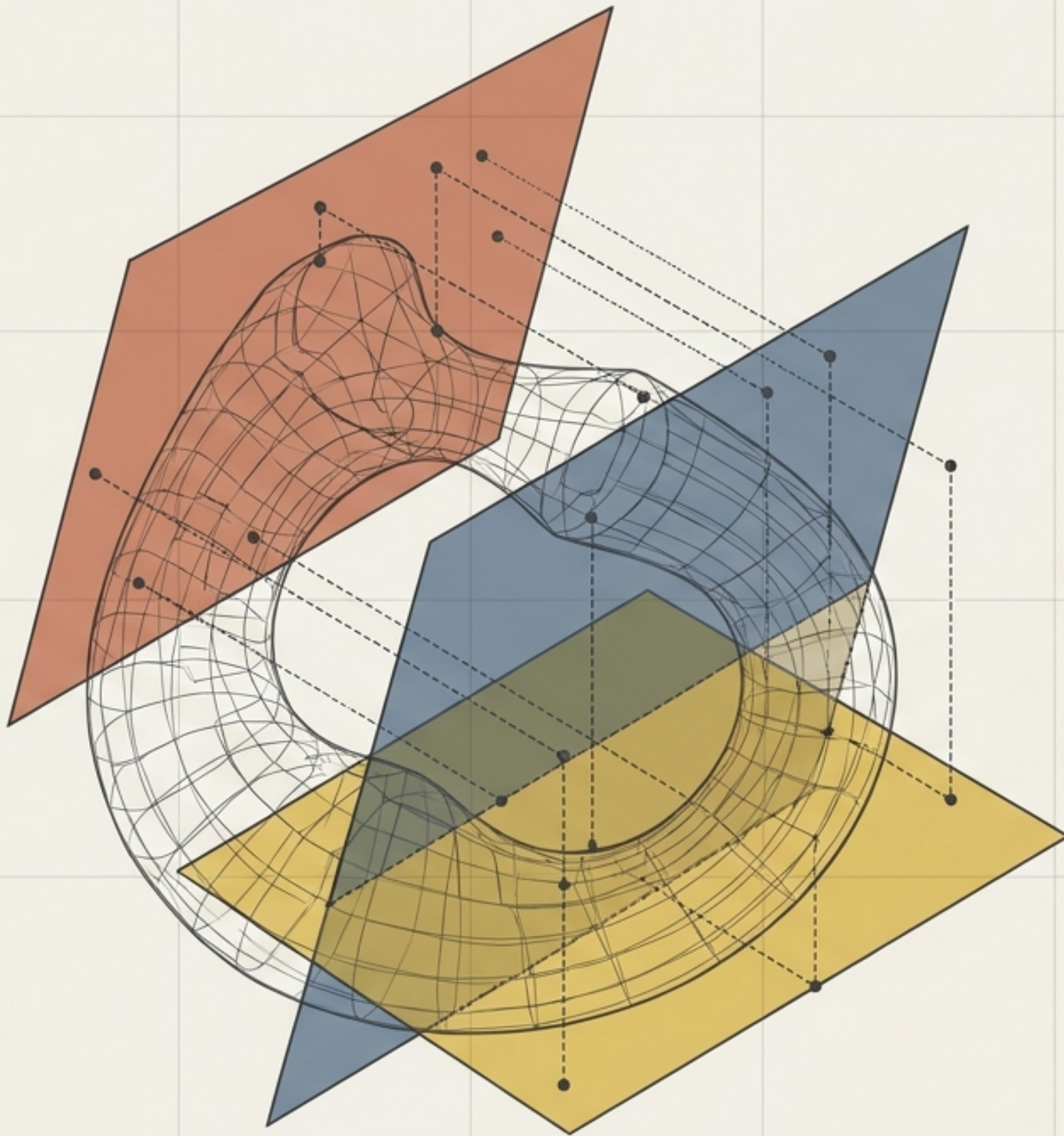




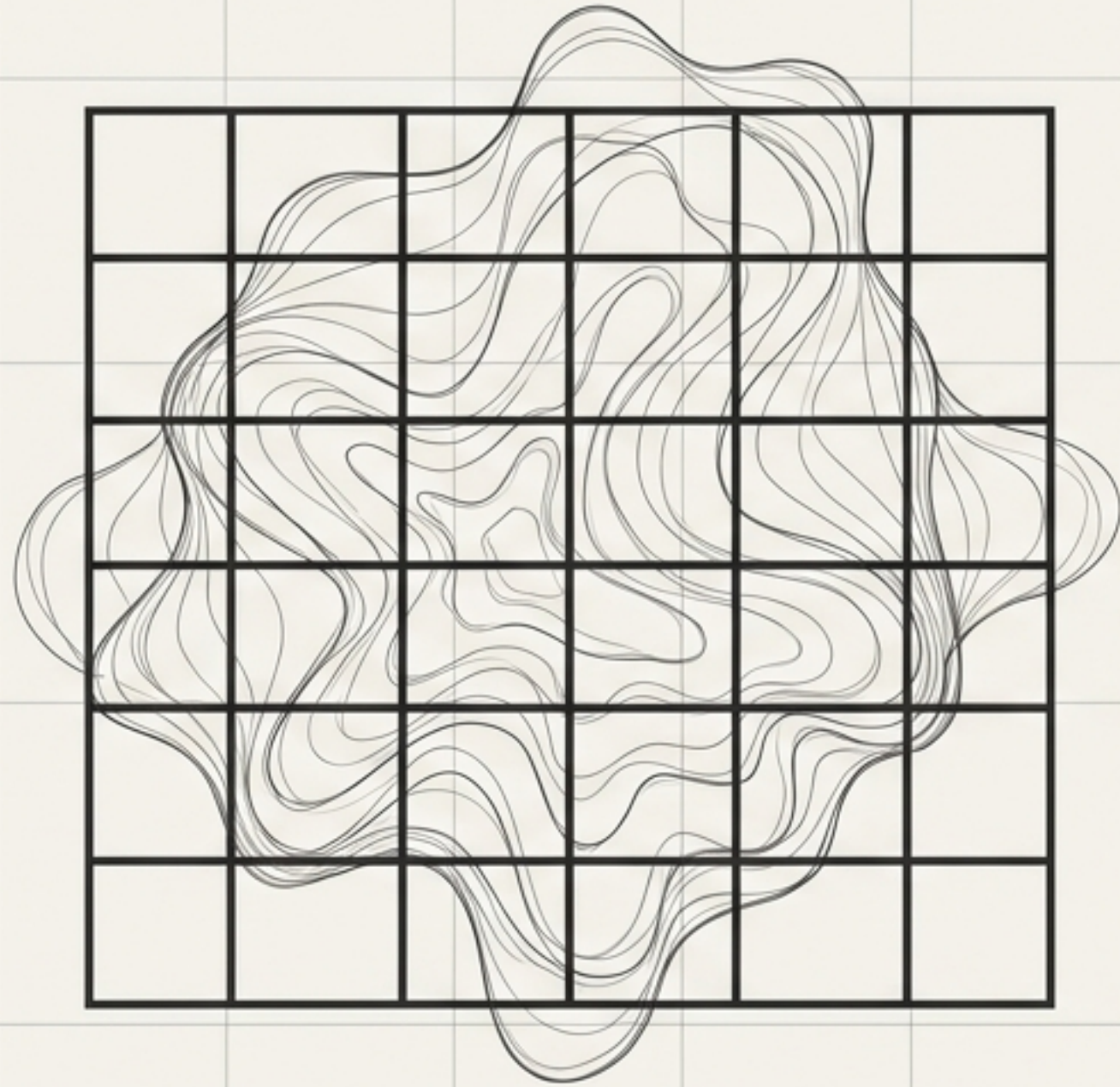
Holocoordinate SystemSystems

Moving Beyond Naive Measurement

Multidifference Observation, Transition Structure,
and Residual Growth in Enriched Atlases.

A coordinate is ordinarily a number attached to a state. This is the math for a richer object: an **atlas of perspectives** that holds itself accountable to reality.

The Illusion of Capture in Applied Settings



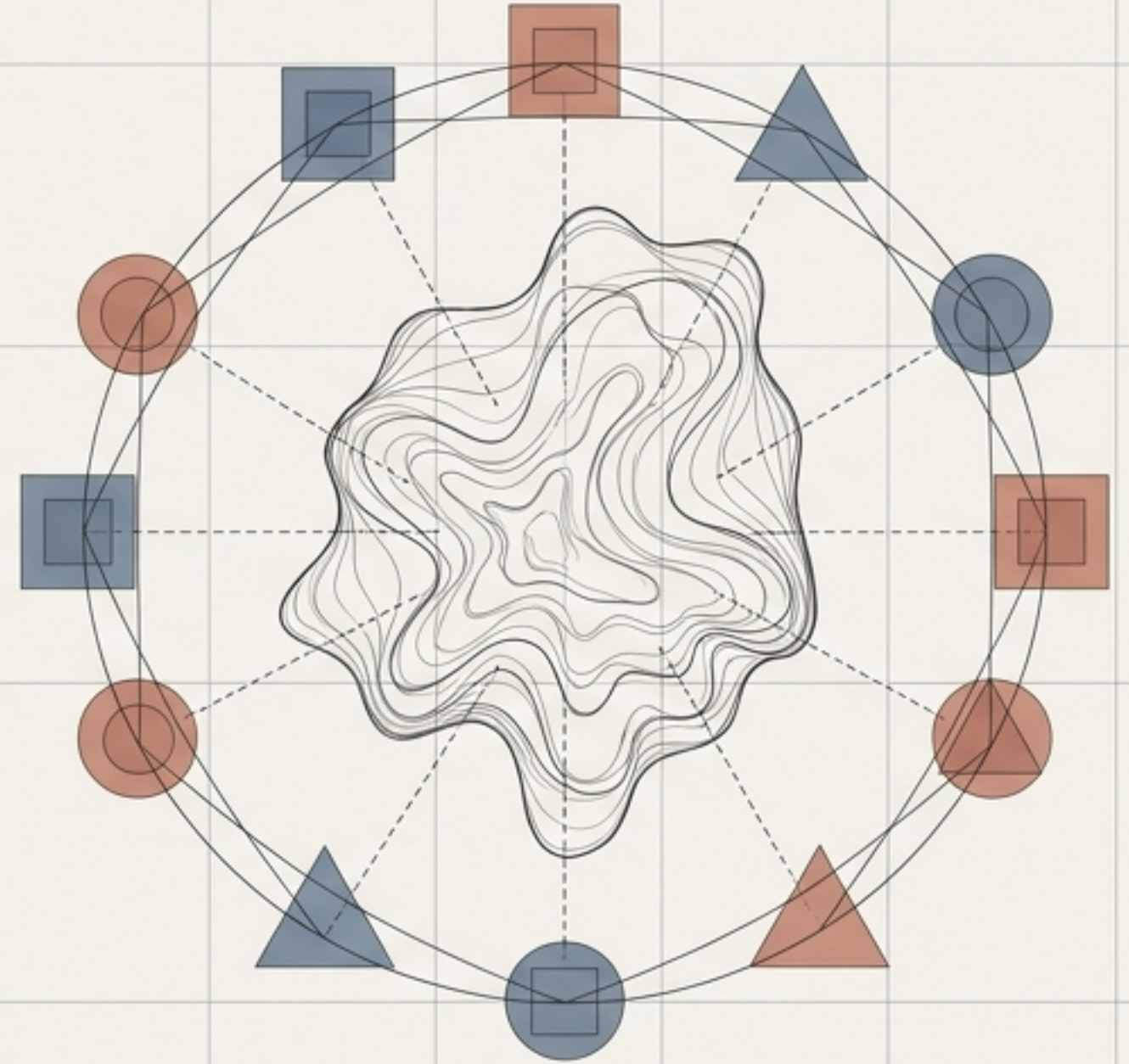
THE COMMON ERROR

We fit parameters within a model and read them as absolute properties of the phenomenon itself.

1. Which state changes leave numbers unchanged?

2. Can numbers from one chart translate to another?

3. What remains unexplained?



THE ENRICHED ATLAS

An interconnected network of adaptable observation nodes.

A true coordinate system must rigorously separate the instrument's limitations from the phenomenon's reality.

The Four Demands of an Enriched Atlas

1. DISCRIMINATION

Local Coordinate: ϕ_α

Registers local differences into feature space V_α . Every nonzero local difference in X must register in at least one perspective.

2. RELATIONALITY

Signature: σ_α

Links the state to the wider field via kernel signatures. Locates a state through broader relations, not just independent axes.

3. TRANSLATION

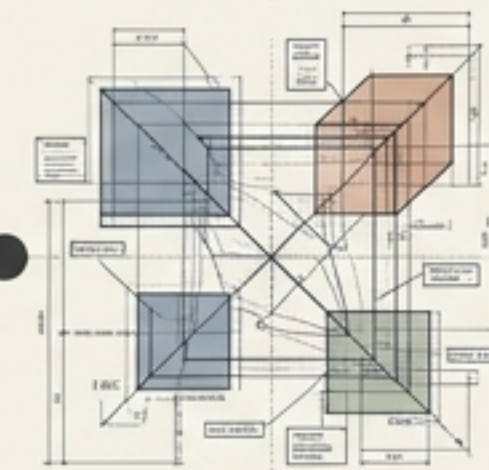
Transition Maps: $T_{\beta\alpha}$

Calculates maps from V_α to V_β . Retains the means to move representations between perspectives to measure path dependence.

4. ACCOUNTABILITY

Residuals: ϵ_α

Extracts the unexplained variance. Records what the chart fails to reproduce to motivate system growth.



Perspective Chart α

Insight: These four demands are formalized into logically independent mathematical properties.

Eliminating the Blind Space

Property 1 – Multidifference Completeness

The Blind Space: $N_\alpha(x)$

The set of tangent directions where motion produces zero first-order change. The chart has no coordinate for these directions.

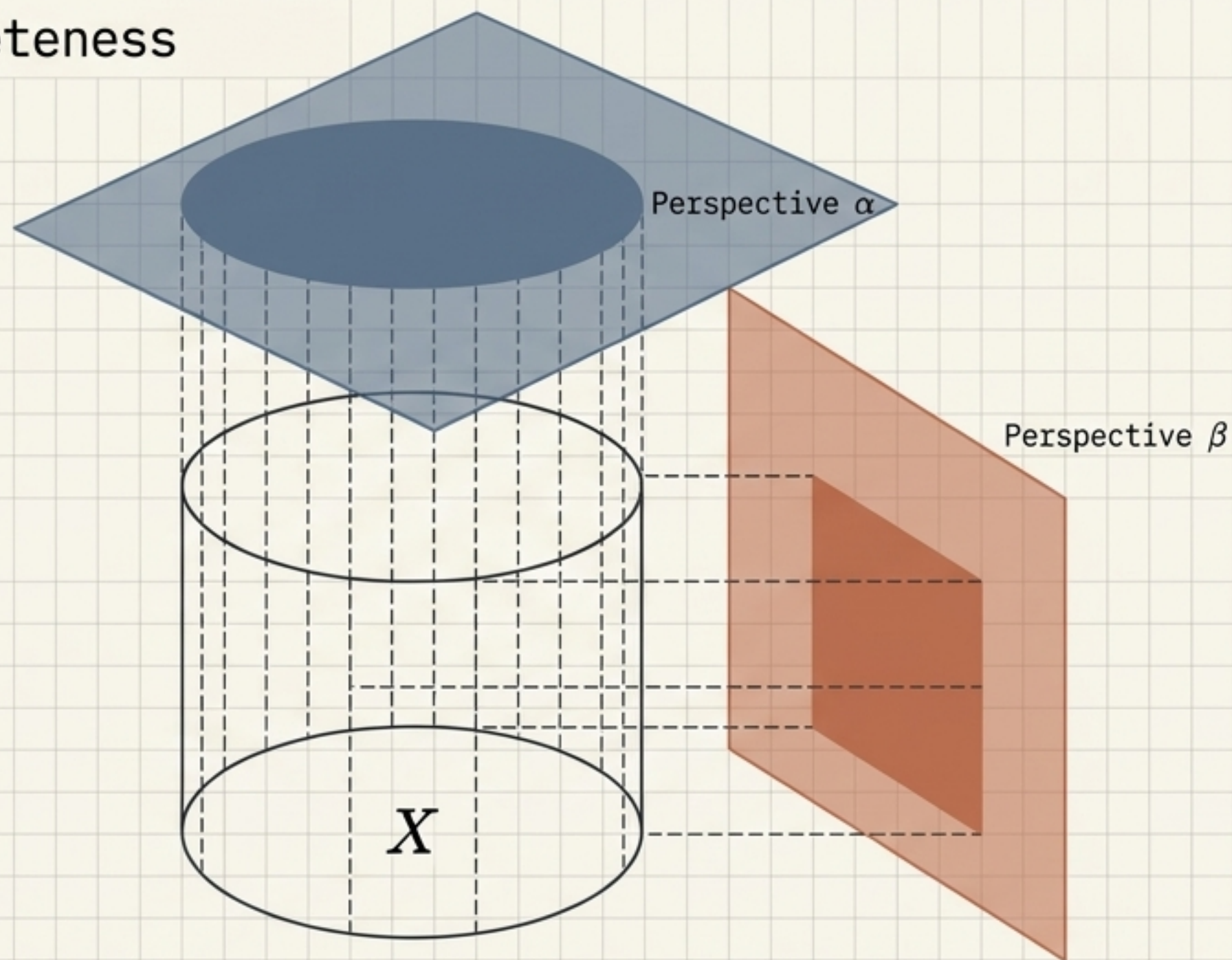
Joint Completeness

The family is complete at x when the joint blind space intersects trivially. The stacked Jacobian $J(x)$ has full rank m .

$$N(x) = \bigcap N_\alpha(x) = \{0\}$$

Error Amplification

Even if complete, observation can be ill-conditioned. If condition number κ_{obs} is high, minor noise amplifies into massive uncertainty.



Conditioning Note: First-order blindness doesn't mean a state is indistinguishable—it indicates degeneracy of observation, a direction technically observable but nearly invisible.

True Holography Requires Injective Signatures

Property 2 – Relationality

The Relational Signature: σ_α

A compressed description of how the wider field appears from x , rather than locating it via independent axes.

$$\sigma(x) = (d(x, \ell_1), \dots, d(x, \ell_N))$$

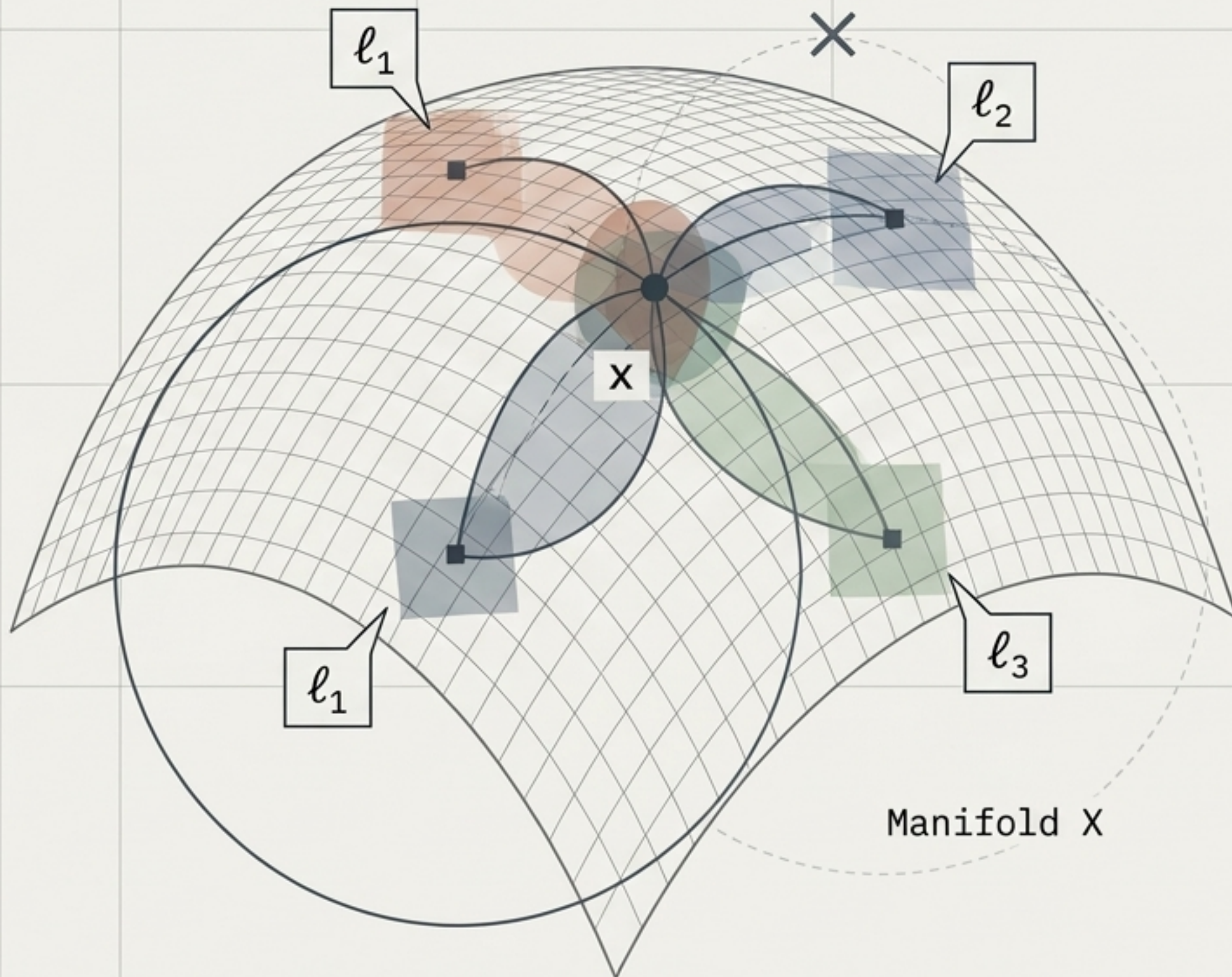
Definition of Strong Holography

Achieved only if the signature guarantees perfect reconstruction of the preserved distinctions.

$$\sigma_\alpha(x) = \sigma_\alpha(y) \Rightarrow x \sim y$$

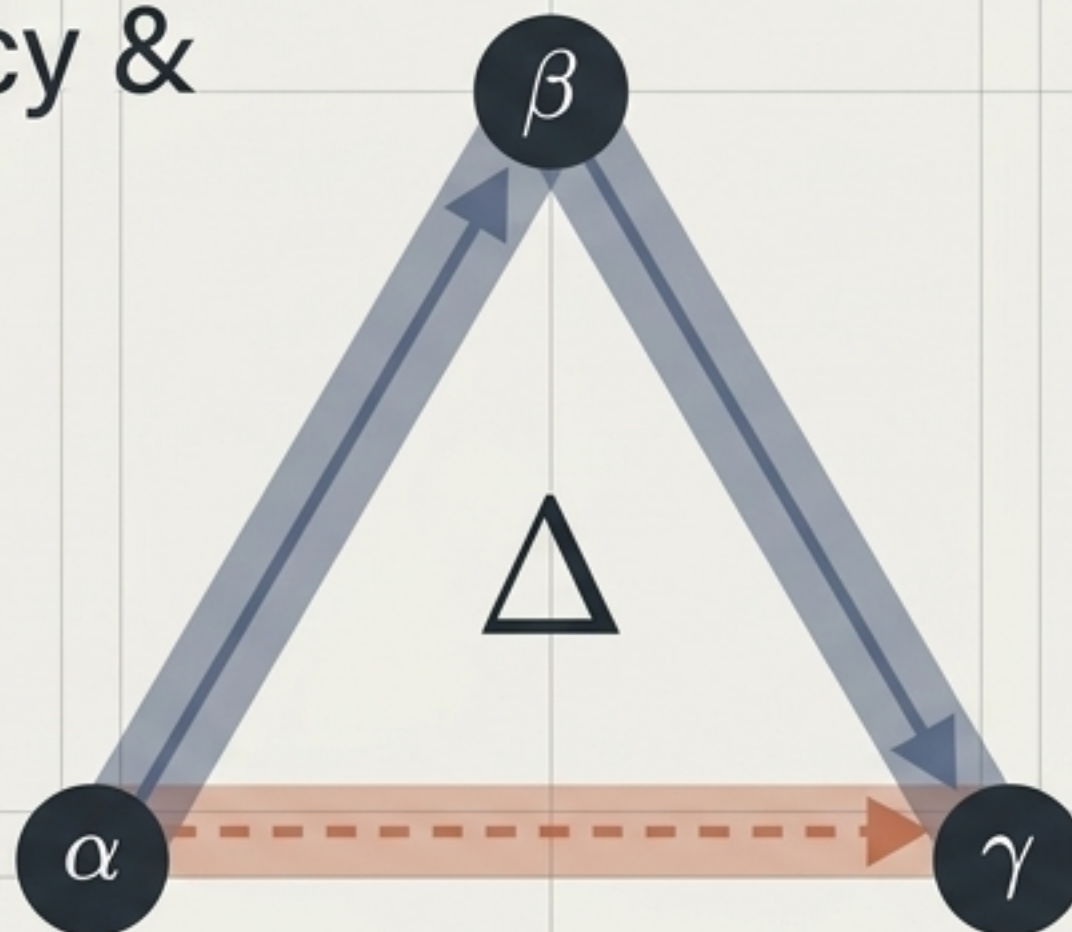
The Trilateration Proof

In Euclidean m -space, exactly $m+1$ affinely independent landmarks guarantee strong holography. Fewer landmarks result in indistinguishable reflections, permanently failing the holography test.



Transition Consistency & The Cocycle Defect

Property 3 — Translation



Definitions

Transition Discrepancy ($\delta_{\alpha\beta}$):

The distance between a direct observation in β and a translation from α . Nonzero discrepancy is not merely noise—it reflects fundamental distortion or missing coordinates.

Cocycle Defect ($\kappa_{\alpha\beta\gamma}$):

The geometric difference between translating through an intermediate chart versus translating directly.

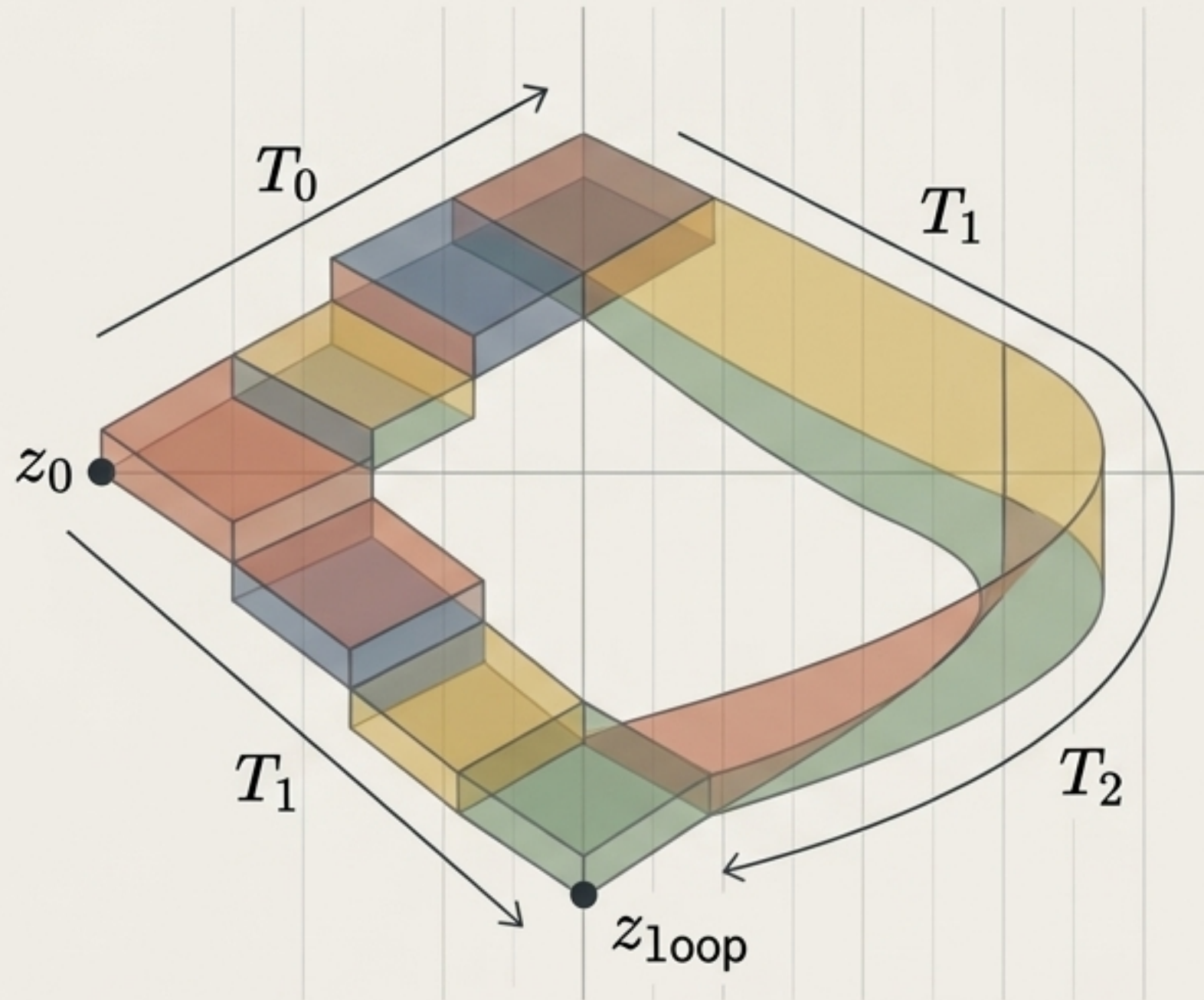
The Bounding Rule

$$\kappa_{\alpha\beta\gamma}(y) \leq L_{\gamma\beta}\delta_{\alpha\beta}(x) + \delta_{\beta\gamma}(x) + \delta_{\alpha\gamma}(x)$$

Rule: If all pairwise discrepancies vanish, the cocycle condition holds perfectly across the transition triangle.

Path Dependence & The Winding Paradox

Property 4 – Holonomy



1 The Holonomy Map

What happens when we transport a representation entirely around a closed loop of perspectives?

$$H_\gamma = T_{k-1} \circ \dots \circ T_1 \circ T_0$$

2 The Accumulation Bound

The global holonomy discrepancy (Ω_γ) is strictly bounded by the accumulated sum of pairwise discrepancies along the path, multiplied by Lipschitz constants.

3 The Winding Paradox

Take a circle mapped by 3 exact overlapping charts. Every chart is a perfect coordinate. Every triple overlap has zero cocycle defect.

Yet, traversing the loop adds 2π to the angle.

Insight: Holonomy is the winding number of the space—invisible to any single chart, and invisible to any triple of charts, yet defining the global reality.

The Architecture is Logically Independent

	Multidifference Complete (M)	Strongly Holographic (H)	Transition-Consistent (C)	Holonomy-Trivial (V)
The Raw Discriminator ($\phi_1=x_1, \phi_2=x_2$) Discriminates every difference, but fails to reconstruct the state.	✓	✗	✓	✓
The Flawed Reconstructor (Shifting transitions) Reconstructs perfectly, but translations are completely inconsistent.	✓	✓	✗	✗
The Winding Loop (The Circle Paradox) Locally exact everywhere, but carries global path-dependence.	✓	✓	✓	✗

THE CORE SYNTHESIS: Independence is the primary value of this framework. A claim that a system is “holographic” warrants signature injectivity—and absolutely nothing else. It guarantees neither consistency nor completeness.

Holding Perspectives to Account

Capturing the Residual

The Mechanism

An enriched atlas does not blindly absorb prediction failure as "noise." It isolates the precise residual difference between reality and the chart's conditional expectation.

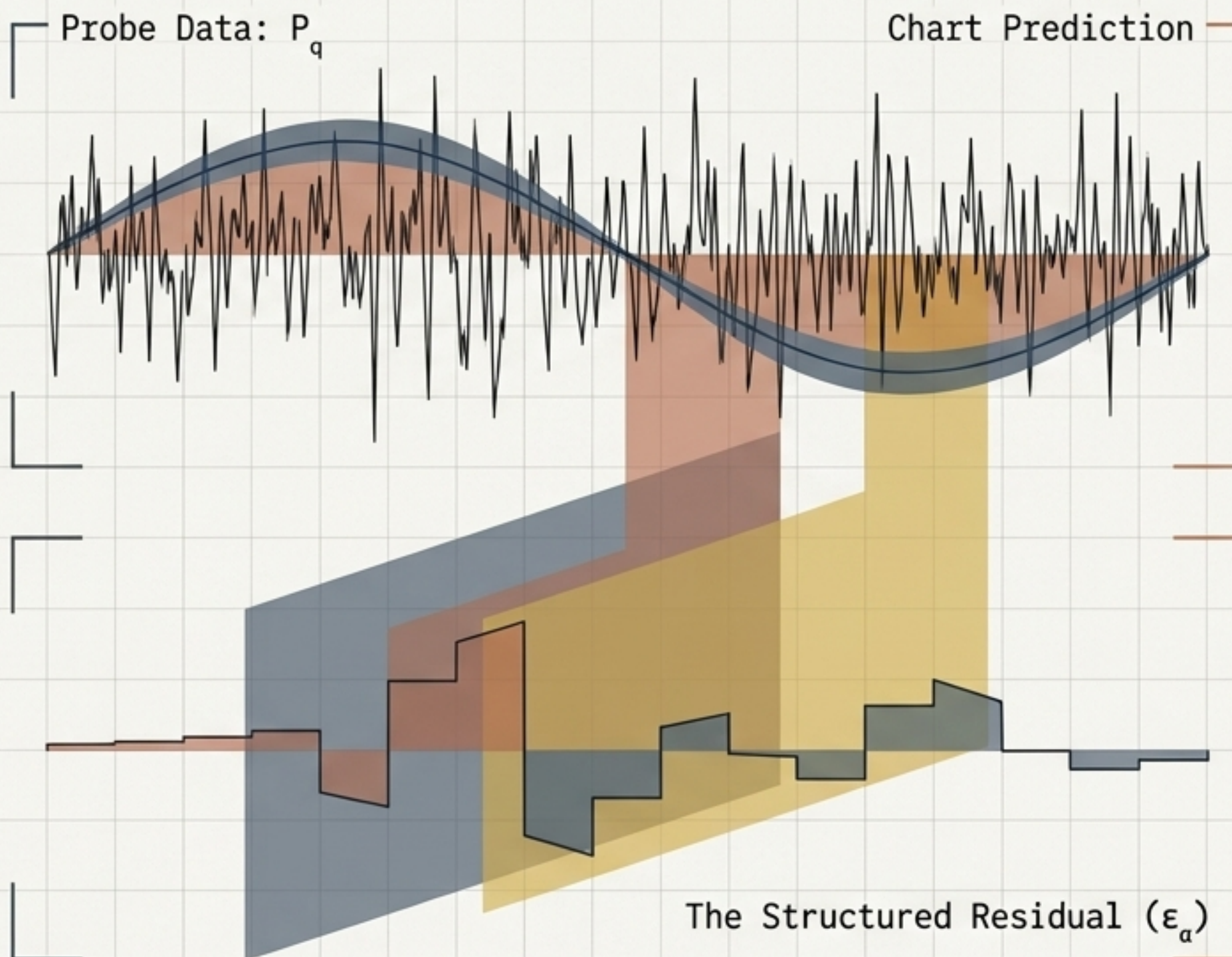
The Math of Reduction

Extracting a chart using the auodens are predicted then residuated pending $\hat{\epsilon}_n$ to chart's expectation:

$$E[\tilde{\epsilon}_{\alpha,q}^2] = E[\epsilon_{\alpha,q}^2] - E[(E[P_q|\phi_{\alpha},z] - E[P_q|\phi_{\alpha}])^2]$$

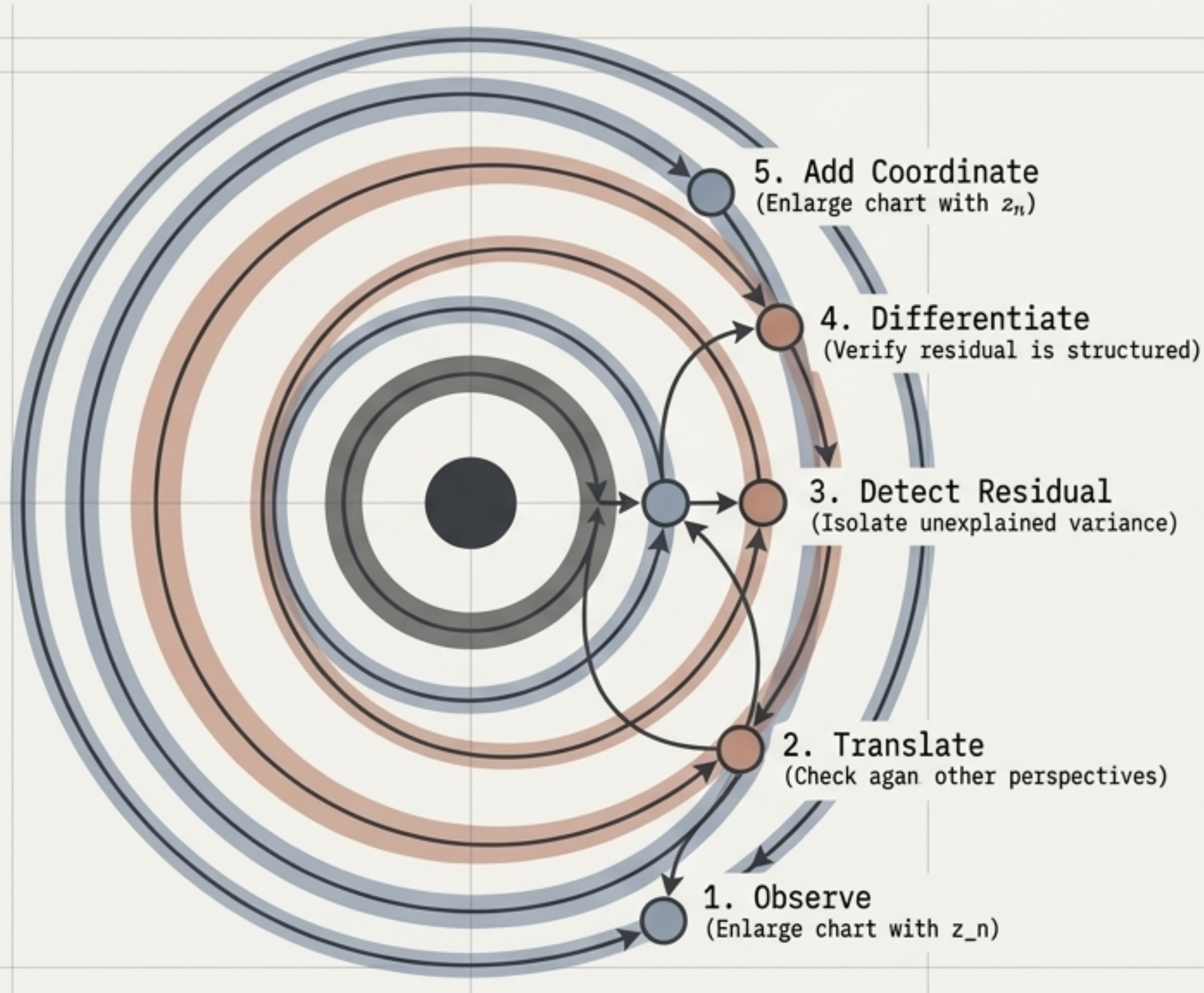
The Rule of Structure

Enlarging a chart never increases mean squared error. But residuals only shrink if they are structurally correlated with respect to the newly proposed coordinate (z). Pure unstructured noise cannot be reduced by arbitrary enlargement.



A Martingale Limit for System Evolution

The Iterative Growth Rule



The Limit Theorem

As coordinates $(z_1, z_2, \dots)$ are successively added, and the sigma-algebra expands, the mean squared residual is monotonically decreasing.

The Math

$$F_n = \sigma(\phi_\alpha, z_1, \dots, z_n)$$

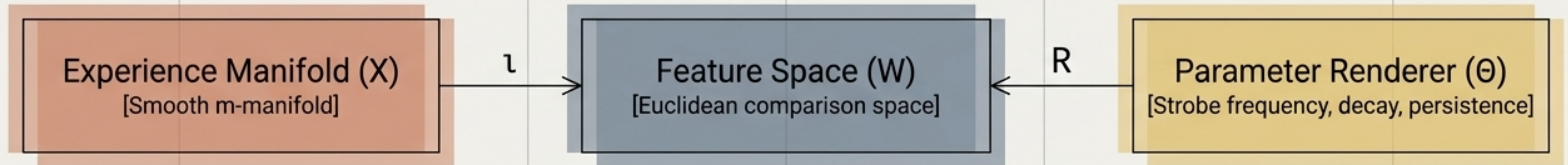
$$\lim_{n \rightarrow \infty} E[\epsilon_n^2] = 0 \text{ iff } P_q \text{ is } F_\infty\text{-measurable}$$

Cautionary Note

Warning: This limit functions strictly at a population level. With finite data, aggressive coordinate enlargement simply fits to noise. Detecting a genuinely structured residual requires held-out probes.

Applying the Atlas to Experiential Data

The Tracer Replication Tool



The Setup

An instrument converts a compressed parameter vector (Θ) into an exposed rendering (W). A participant compares this rendering against their internal subjective experience (X).

Assumption: Comparison is mediated by squared distance: $D_e(w) = ||\iota(e) - w||^2$

The Tool Chart (ϕ_τ)

The tool itself operates as a localized coordinate chart. The mapped coordinate is mathematically defined as the inverse of the renderer applied to the nearest projection of the experience.

$$\phi_\tau(e) = R^{-1}(\pi(\iota(e)))$$

Instrument Blind Spaces & Branch Dependencies

The Limits of Parameterization

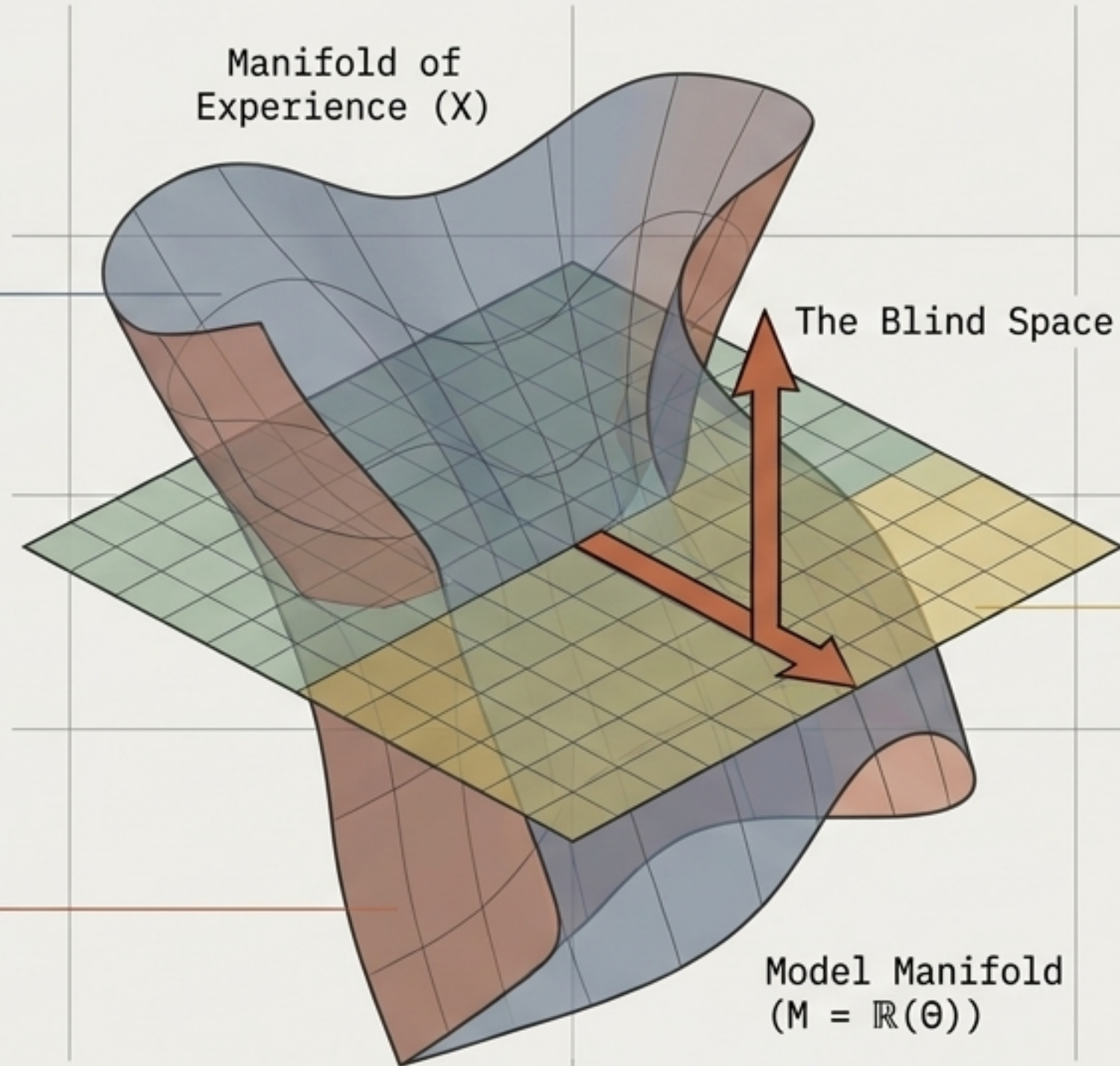
The Blind Space

The variation for which the parameterization has no coordinate. It comprises the set of changes in experience whose perceptual image is exactly normal (perpendicular) to the model manifold.

The Dimensional Trap

If the dimension of the experience (m) exceeds the tool parameters (p), the tool mathematically cannot be complete on its own.

$$\dim N_{\tau}(\mathbf{e}) \geq m - p$$

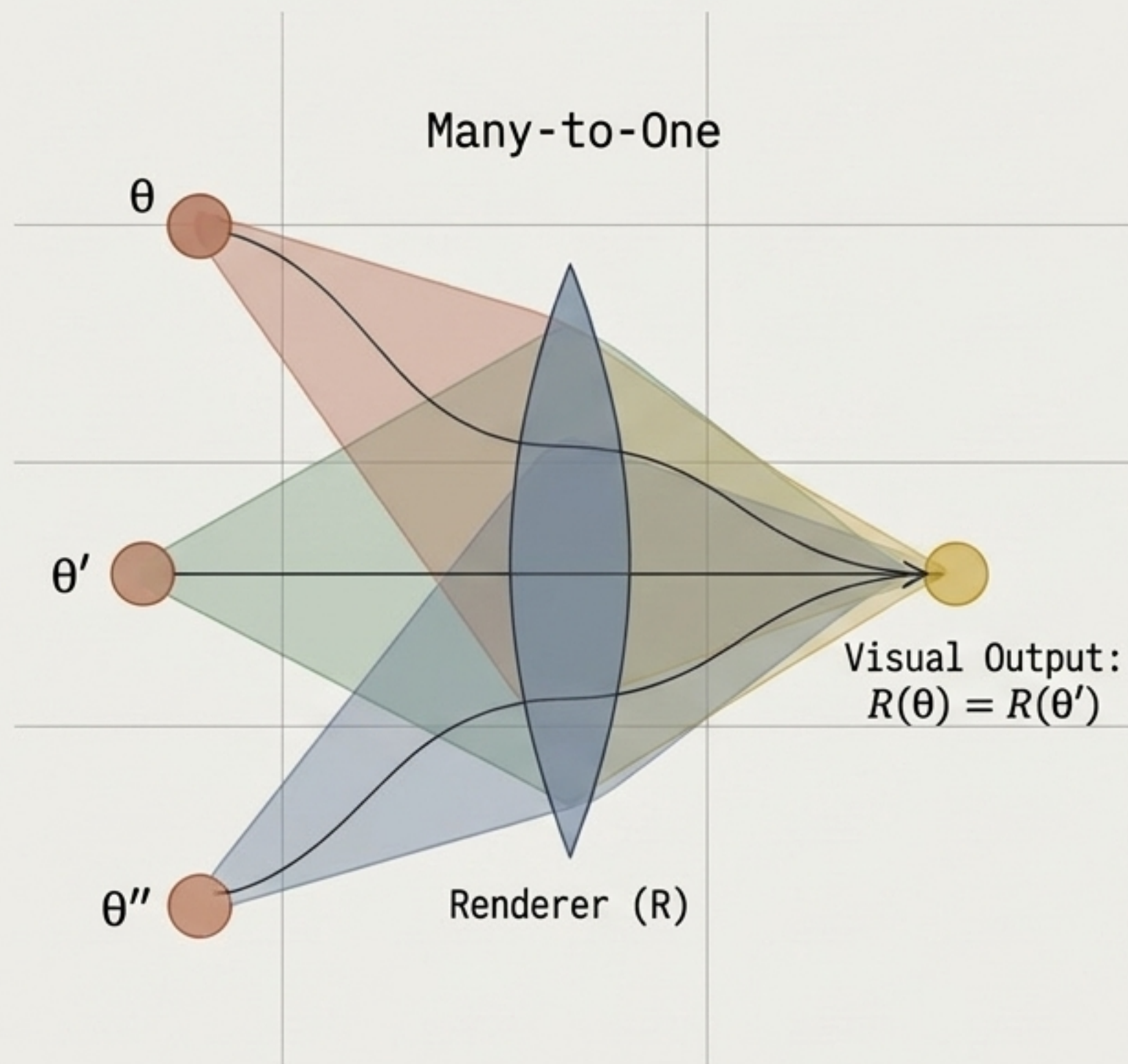


Branch Dependence

If the renderer is an immersion but not injective, the instrument generates overlapping geometric "sheets." The fitted parameter becomes multi-valued, and the blind space becomes a property of the specific branch chosen by the user.

When the Instrument is Degenerate

Parameter Redundancy & Gauge Choice



Parameter Redundancy

When the instrument rank drops, distinct parameter vectors produce physically indistinguishable rendering outputs.

$$\theta \sim_R \theta'$$

Well-Defined Functions

A derived metric (like a fitted replay frequency) is only mathematically valid if it remains totally constant across all redundant parameters.

If $R(\theta) = R(\theta')$ but $f(\theta) \neq f(\theta')$, the metric is invalid.

Holonomy as Gauge

If we render a parameter, re-fit it, and get a discrepancy, this is a coordinate artifact. It vanishes entirely if the chart is treated mathematically as a quotient space $(\theta/\sim_R)$ rather than linear Euclidean axes.

Criteria for Coordinate Transport

Proving biological and physical correspondence.

The Tool
Chart
(ϕ_T)
[Experiential
Parameter]

1. Low Discrepancy
The map predicts the target on held-out states (proving it isn't just overfitted).

2. Low Cocycle Defect
The correspondence holds true even when verified against independent
3rd-party perspectives (e.g., psychophysics).

3. Joint Completeness
The agreement isn't secretly hiding within
a dimensional blind space.

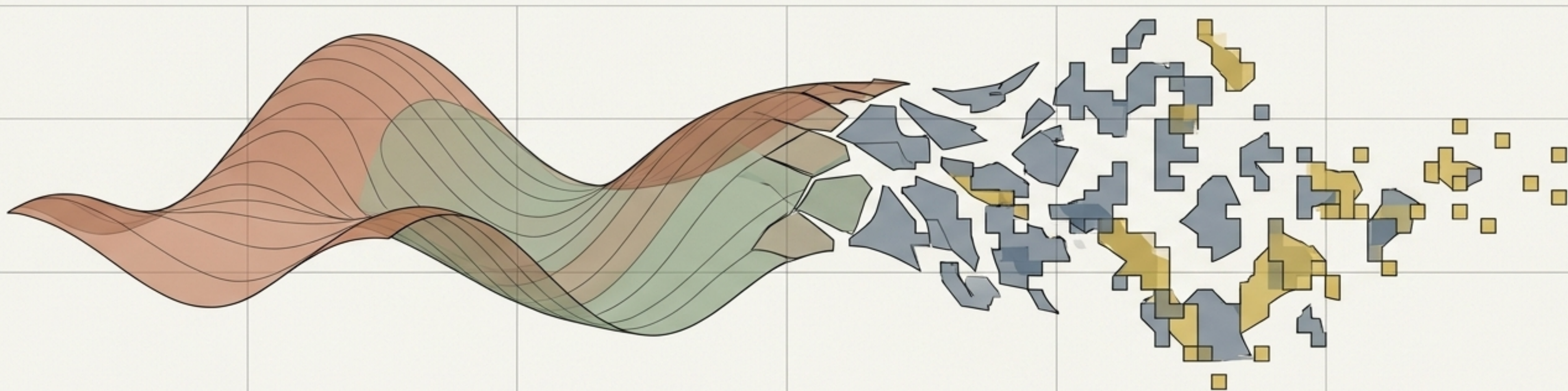
4. Well-Defined Quotient
The transported quantity mathematically survives
all parameter redundancy tests.

The Neural
Target
(ϕ_V)
[e.g., fMRI
Feature Vectors]

Conclusion: Failing these criteria means an instrument only proves
resemblance within the renderer, not a true biological correspondence.

Where the Mathematics Bite

Scope, Limits, & Boundary Constraints



1. The Smoothness Hypothesis

This differential geometry heavily assumes experiential state spaces are smooth manifolds. For observational data, this is a profound modeling hypothesis, not an empirical finding.

2. Finite Data Reality

True derivatives ($D\phi_\alpha(\mathbf{x})$) cannot be observed from finite reports. They must be approximated by the numerical rank of estimated Jacobians, introducing significant stability caveats.

3. Local vs. Global

Completeness is strictly a local, infinitesimal property. It does not guarantee global injectivity, nor does it guarantee that the perspective spans the actual phenomenon rather than just the instrument.